

Chapter 9:

Infinite Series

Section 9.9: Representation of Functions by Power Series

Geometric Power Series

EXAMPLE

Find the sum of the series $\sum_{n=0}^{\infty} x^n$, where $|x| < 1$.

SOLUTION Notice that this series starts with $n = 0$ and so the first term is $x^0 = 1$. (With series, we adopt the convention that $x^0 = 1$ even when $x = 0$.) Thus

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \cdots$$

This is a geometric series with $a = 1$ and $r = x$. Since $|r| = |x| < 1$, it converges and

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}$$

■

We start with an equation that we have seen before.

1

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n \qquad |x| < 1$$

A geometric illustration of Equation 1 is shown in Figure 1. Because the sum of a series is the limit of the sequence of partial sums, we have

$$\frac{1}{1-x} = \lim_{n \to \infty} s_n(x)$$

where

$$s_n(x) = 1 + x + x^2 + \cdots + x^n$$

is the nth partial sum. Notice that as n increases, $s_n(x)$ becomes a better approximation to $f(x)$ for $-1 < x < 1$.

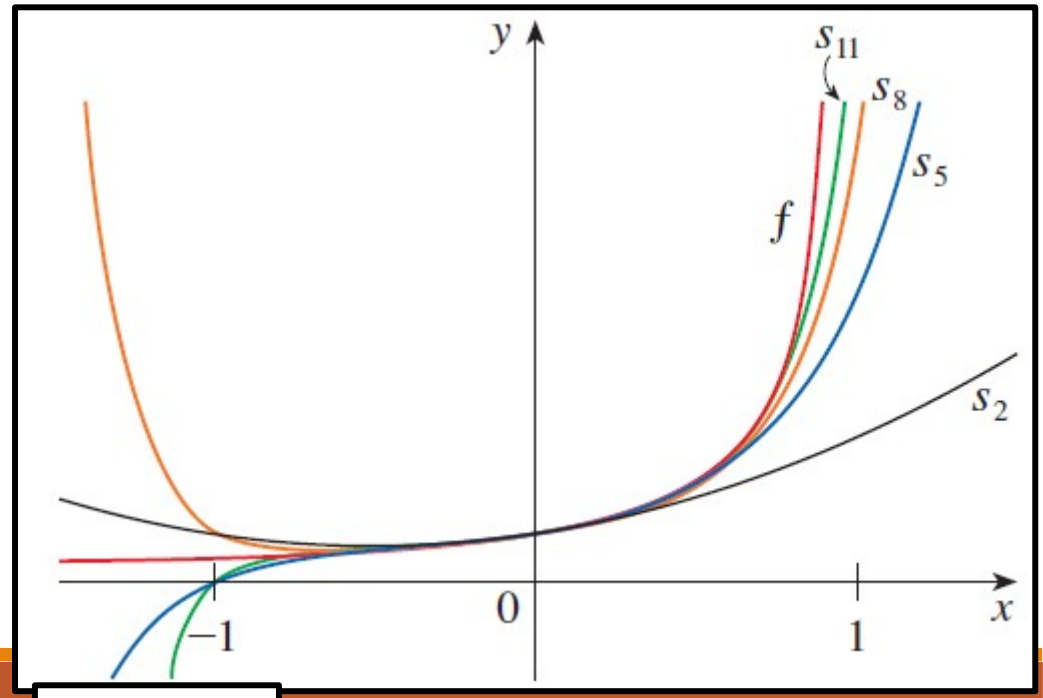


FIGURE 1

EXAMPLE

Find a power series representation for the following function centered at 0.

$$f(x) = \frac{1}{1+x}$$

SOLUTION

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

EXAMPLE

Find a power series representation for the following function centered at 0.

$$f(x) = \frac{4}{x+2}$$

SOLUTION

$$\frac{4}{x+2} = \sum_{n=0}^{\infty} 2\left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n-1}}, \qquad |x| < 2$$

EXAMPLE

Find a power series representation for the following function centered at 0.

$$f(x) = \frac{1}{5 + 3x}$$

SOLUTION

$$\frac{1}{5+3x} = \sum_{n=0}^{\infty} \frac{(-1)^n 3^n \, x^n}{5^{n+1}} \,, \qquad |x| < \frac{5}{3}$$

EXAMPLE

Find a power series representation for the following function centered at 1.

$$f(x) = \frac{1}{x}$$

SOLUTION

$$\frac{1}{x} = \sum_{n=0}^{\infty} (-1)^n (x-1)^n \qquad |x-1| < 1$$

Operations with Power Series

The versatility of geometric power series will be shown later in this section, following a discussion of power series operations. These operations, used with differentiation and integration, provide a means of developing power series for a variety of elementary functions. (For simplicity, the operations are stated for a series centered at 0.)

Operations with Power Series

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ and $g(x) = \sum_{n=0}^{\infty} b_n x^n$.

1. $f(kx) = \sum_{n=0}^{\infty} a_n k^n x^n$

2. $f(x^N) = \sum_{n=0}^{\infty} a_n x^{nN}$

3. $f(x) \pm g(x) = \sum_{n=0}^{\infty} (a_n \pm b_n) x^n$

Operations with Power Series

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The operations described above can change the interval of convergence for the resulting series. For example, in the addition shown below, the interval of convergence for the sum is the *intersection* of the intervals of convergence of the two original series.

$$\underbrace{\sum_{n=0}^{\infty} x^n}_{(-1,1)} + \underbrace{\sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n}_{(-2,2)} = \underbrace{\sum_{n=0}^{\infty} \left(1 + \frac{1}{2^n}\right) x^n}_{(-1,1)}$$

$(-1, 1) \cap (-2, 2) = (-1, 1)$

EXAMPLE

Find a power series representation for the following function centered at 0.

$$f(x) = \frac{3x - 1}{x^2 - 1}$$

SOLUTION

$$\frac{3x-1}{x^2-1} = \sum_{n=0}^{\infty} [2(-1)^n - 1]\, x^n$$

The interval of convergence for this power series is $(-1, 1)$.

EXAMPLE

Using a Power Series **In Exercises 19–28, use the power series**

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

to find a power series for the function, centered at 0, and determine the interval of convergence.

25. $g(x) = \dfrac{1}{x^2 + 1}$

SOLUTION

$$\frac{1}{x^2+1} = \sum_{n=0}^{\infty} (-1)^n \left(x^2\right)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}, (-1, 1)$$

EXAMPLE 4 Finding a Power Series by Integration

Find a power series for

$$f(x) = \ln x$$

centered at 1.

SOLUTION

$$\ln x = \sum_{n=0}^{\infty} (-1)^n \frac{(x-1)^{n+1}}{n+1}$$

Interval of convergence: $(0, 2]$

Remark About the number ln 2

EXAMPLE

Find a power series for

$$g(x) = \arctan x$$

centered at 0.

SOLUTION

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

Interval of convergence: $(-1, 1)$

Remark About the number π

EXAMPLE

Approximating a Value In Exercises 33–36, use the power series for $f(x) = \arctan x$ to approximate the value, using $R_N \le 0.001$.

35. $\displaystyle\int_0^{1/2} \frac{\arctan x^2}{x}\,dx$

SOLUTION

$$\int_0^{1/2} \frac{\arctan x^2}{x}\,dx = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(4n+2)(2n+1)2^{4n+2}} = \frac{1}{8} - \frac{1}{1152} + \cdots$$

$$\int_0^{1/2} \frac{\arctan x^2}{x}\,dx \approx 0.125.$$

[1] 0s
```
from sympy import *
x=symbols("x")
```

[2] 12s
```
integrate(atan(x**2)/x,x).doit()
```

$$\int \frac{\operatorname{atan}\left(x^2\right)}{x}\,dx$$

[3] 5s
```
integrate(atan(x**2)/x,(x,0,0.5))
```

$$\int_{0}^{0.5} \frac{\operatorname{atan}\left(x^2\right)}{x}\,dx$$

[4] 7s
```
integrate(atan(x**2)/x,(x,0,0.5)).doit()
```

$$\int_{0}^{0.5} \frac{\operatorname{atan}\left(x^2\right)}{x}\,dx$$

[5] 5s
```
I=float(integrate(atan(x**2)/x,(x,0,0.5)))
I
```

```
0.12415087549115343
```

[6] 0s
```
float(1/8) # we need only the first term of the power series of the integral
```

```
0.125
```

0s
```
(1/8)-I # Error estimation
```

```
0.0008491245088465688
```

EXAMPLE

Using a Power Series **In Exercises 37–40, use the power series**

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

to find a power series for the function, centered at 0, and determine the interval of convergence.

37. $f(x) = \dfrac{1}{(1-x)^2}$

SOLUTION

$$\frac{1}{(1-x)^2} = \frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{d}{dx}\left[\sum_{n=0}^{\infty} x^n\right] = \sum_{n=1}^{\infty} nx^{n-1}, \; |x| < 1$$

EXAMPLE

Think About it

Using a Power Series In Exercises 37–40, use the power series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

to find a power series for the function, centered at 0, and determine the interval of convergence.

38. $f(x) = \dfrac{x}{(1-x)^2}$

SOLUTION

$$\frac{x}{(1-x)^2} = x\sum_{n=1}^{\infty} nx^{n-1} = \sum_{n=1}^{\infty} nx^n, |x| < 1$$

EXAMPLE

Using a Power Series **In Exercises 19–28, use the power series**

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

to find a power series for the function, centered at 0, and determine the interval of convergence.

23. $f(x) = \ln(x+1) = \int \frac{1}{x+1}\,dx$

SOLUTION

$$\ln(x+1) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}, (-1, 1].$$

EXAMPLE

Finding the Sum of a Series **In Exercises 47–52, find the sum of the convergent series by using a well-known function. Identify the function and explain how you obtained the sum.**

48. $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{3^n n}$

SOLUTION

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{3^n n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(1/3)^n}{n} = \ln\left(\frac{1}{3} + 1\right) = \ln \frac{4}{3} \approx 0.2877.$$

Solve exercises of section 9.9.

Start with the suggested problems mentioned in the syllabus.